

Data Flow Diagram

Date	1 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Farmer/User who enters soil and environmental parameters into the system and receives the recommended crop as output.
Rectangle with numbered header	Process	Performs the main operations of the system such as Input Validation , Machine Learning Prediction , and Displaying the Crop Recommendation .
Rectangle (solid fill, no number)	Data Store	Stores the trained machine learning model (crop_model.pkl) and the Crop Recommendation Dataset (CSV) used to generate crop predictions.
Labeled arrow	Data Flow	Represents the movement of data between the user, processes, and data store. Examples include Soil Parameters , Validated Input , Prediction Request , Predicted Crop , and Crop Recommendation Result .

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

DATA FLOW DIAGRAM – OPTICROP (CROP RECOMMENDATION SYSTEM)

